

GRADUATE PROJECT REPORT
Operations Research / Business Analytics

Minimizing Peak Post-Match Queues
Optimal Bus Dispatch Scheduling to Support Stadium Egress
at MetLife Stadium During FIFA World Cup 2026

VENUE	MetLife Stadium, East Rutherford, New Jersey
EVENT	FIFA World Cup 2026 — Final: July 19, 2026
MODEL	Mixed-Integer Linear Program (MILP)
OBJECTIVE	Minimize peak post-match passenger queue (z^*)
BASE CASE	24,000 pass. 60 buses $\tau=60$ min $z^*=6,490$

TEAM

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1. Executive Summary

Planning Horizon 3 Hours <i>12 × 15-min intervals</i>	Total Demand (Base) 24,000 <i>outbound passengers</i>	Peak Queue (Base) 6,490 <i>passengers at interval 5</i>	Queue Cleared By Interval 11 <i>within planning horizon</i>
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MetLife Stadium in East Rutherford, New Jersey is scheduled to host eight FIFA World Cup 2026 matches, including the final on July 19, 2026. Post-match crowd egress represents one of the most pressing operational challenges for transit planners. When the final whistle sounds, a large proportion of the stadium’s outbound passengers attempt to leave within a narrow time window, creating a sharp surge in demand that fixed infrastructure may not be able to absorb quickly enough.

This report presents a time-indexed Mixed-Integer Linear Programming (MILP) model designed to minimize the maximum passenger queue during the post-match egress period. The model focuses on the shuttle bus operation between MetLife Stadium and Secaucus Junction, with bus dispatch decisions optimized across 12 consecutive 15-minute intervals spanning a 3-hour planning horizon.

Project Deliverable Summary

The base case achieves an optimal peak queue of 6,490 passengers (interval 5). Demand sensitivity analysis spans 18,000–32,000 passengers, with peak queues ranging from 2,660 to 12,520. An operational capacity ceiling is identified at 26,000 passengers. One-at-a-time (OAT) sensitivity analysis isolates the independent effect of each parameter. Four key managerial recommendations are issued: (1) front-load bus dispatch; (2) reduce turnaround time; (3) manage demand timing; (4) plan contingencies above the capacity ceiling.

2. Project Background & Problem Statement

2.1 Operational Context

MetLife Stadium will serve as one of the primary venues for the 2026 FIFA World Cup, with a soccer capacity of 82,500 spectators and eight scheduled matches including the Final. NJ TRANSIT has identified the primary public transit pathway as travel through Secaucus Junction followed by the Meadowlands Rail Line, with service reserved exclusively for match ticketholders on event days.

The critical operational bottleneck is post-match egress. When a match ends, a disproportionately large share of the outbound crowd seeks transportation within a short window, producing a sharp surge in demand that must be served by a combination of fixed rail capacity and supplementary bus service. During the 2025 FIFA Club World Cup, NJ TRANSIT operated more than 150,000 combined bus and rail trips across nine matches, serving 20,390 outbound passengers on the final Sunday. NJ TRANSIT subsequently approved auxiliary bus service support for the 2026 World Cup, explicitly describing the Club World Cup as a test run for the larger tournament.

2.2 Problem Statement & Project Scope

Problem Statement

How should NJ TRANSIT optimally schedule shuttle bus dispatches between MetLife Stadium and Secaucus Junction during the post-match egress window to minimize the worst-case passenger queue, subject to fleet size, turnaround time, and staging capacity constraints?

This project intentionally focuses on one specific bottleneck: the post-match bus dispatch operation from MetLife Stadium to Secaucus Junction. The following are explicitly out of scope:

- Full New York–New Jersey regional transit network modelling
- Rail station platform capacity at Secaucus Junction
- Downstream connections beyond Secaucus
- Private car, rideshare, and charter bus modes

Three real operational constraints are incorporated: effective (not nominal) bus capacity, turnaround time constraints, and staging/loading limits that cap the number of buses that can depart in any single 15-minute interval.

2.3 Why This Problem Merits Analytical Investment

Post-match queue management at MetLife represents a class of operational challenge ideally suited to operations research: demand changes rapidly over time, capacity is constrained by both infrastructure and logistics, and decisions must be made across a finite sequence of short intervals. Furthermore, transportation failures during a globally visible mega-event carry significant reputational and safety consequences, raising the value of pre-event optimization.

From a project management perspective, the compressed planning timeline for a mega-event like the World Cup means that planners cannot rely on trial-and-error. Quantitative decision support tools such as this MILP model allow planners to stress-test operating assumptions, identify binding constraints, and develop contingency plans before game day.

3. Analytical Model — Formulation & Design

The post-match egress problem is formulated as a time-indexed Mixed-Integer Linear Programming (MILP) model. The model tracks the evolution of the passenger queue across 12 consecutive 15-minute intervals, optimizing bus dispatch decisions to minimize the worst-case queue at any point during the planning horizon.

3.1 Model Architecture

Component	Definition
Index	$t = 1, 2, \dots, 12$: post-match 15-minute intervals (3-hour horizon)
Decision Variable: x_t	Number of buses dispatched in interval t (integer ≥ 0)
Decision Variable: y_t	Total passengers served in interval t (continuous ≥ 0)
Decision Variable: q_t	Passenger queue remaining at end of interval t (continuous ≥ 0)

Decision Variable: z	Maximum passenger queue across all intervals (continuous ≥ 0) — the objective
Fixed Input: D_t	Outbound passenger demand arriving in interval t
Fixed Input: R_t	Rail capacity available in interval t (passengers per interval)
Parameters	B = fleet size; C^b = effective bus capacity; τ = turnaround time; L = loading limit; Q_0 = initial queue

3.2 Objective Function

The primary objective is to minimize the maximum passenger queue over all post-match intervals — a minimax formulation linearized by introducing auxiliary variable z :

Minimize $M \cdot z + \sum q_t$

The large weight M on z ensures the minimax objective is never sacrificed in favour of reducing total queue. This lexicographic structure preserves the crowd-safety focus while avoiding unnecessarily large total waiting burdens after the peak has been controlled.

Why Minimax?

A minimax objective is chosen rather than average-queue minimisation because the most serious operational failure mode is not high average waiting time, but extreme instantaneous congestion at a specific point in time. A brief queue spike — even if short-lived — creates crowd-management conditions that are difficult to control, potentially dangerous, and highly visible. The minimax formulation directly targets the worst-case interval.

3.3 Constraints

Constraint	Expression	Operational Meaning
Queue Evolution	$q_t = q_{t-1} + D_t - y_t$	Unmet demand in any interval carries forward to the next
Service Capacity	$y_t \leq R_t + C^b \cdot x_t$	Passengers served cannot exceed rail + bus capacity
Demand Availability	$y_t \leq q_{t-1} + D_t$	Cannot serve phantom demand not yet present in the system
Peak Queue Definition	$z \geq q_t$ for all t	z is exactly the worst-case queue across all intervals
Bus Loading Limit	$x_t \leq L$	Staging and loading capacity caps per-interval dispatch

Fleet Turnaround	$\sum x_k \leq B$ (rolling window h intervals)	Buses in transit or recently dispatched cannot be reused
Integrality	$x_t \in \mathbb{Z}_{\geq 0}; y_t, q_t, z \geq 0$	Fractional buses are operationally meaningless

The fleet turnaround constraint (Constraint 6) is the key mechanism modelling real operational dynamics. It prevents the model from dispatching more buses than are physically available given the time buses spend in transit and returning — a binding factor that planners who simply count total fleet size will systematically overlook.

4. Parameter Estimation & Assumptions

All model parameters are drawn from one of two sources: verified public information about the event setting and NJ TRANSIT operations, or clearly labelled scenario-based planning assumptions for inputs that are not publicly reported at the required level of detail. Readers should treat all results as conditional on these assumptions.

Parameter	Base Case	Source Type	Note
Planning Horizon	3 hrs / 12 intervals	Verified (NJ TRANSIT)	Consistent with published post-match service duration
Total Outbound Demand	24,000 passengers	Scenario assumption	Calibrated to 2025 Club World Cup volumes; scaled for WC
Rail Capacity (R_t)	1,300 pass./interval	Scenario assumption	Estimated from published NJ TRANSIT multilevel car data
Supporting Buses (B)	60 buses	Scenario assumption	MetLife–Secaucus shuttle only; not total WC auxiliary fleet
Bus Capacity (C^b)	55 pass./bus	Scenario assumption	Effective (not nominal) capacity
Turnaround Time (τ)	60 min ($h=4$ intervals)	Scenario assumption	Includes transit, dwell, and return to staging area
Loading Limit (L)	18 buses/interval	Scenario assumption	Physical staging and loading area capacity near stadium

Assumption Transparency

No public information was identified on the specific number of buses assigned to the MetLife-to-Secaucus shuttle, effective capacity per bus, typical turnaround time, or per-interval loading limit. These four parameters (B , C^b , τ , L) are all scenario-based planning assumptions. Demand profiles are also estimated, not empirically derived from interval-level egress data. The scenario and sensitivity analyses in Sections 6–8 are designed to show how model outputs change as these assumptions vary.

5. Base Case Results — Optimal Solution

The base case optimization is solved to optimality. Under base case parameters, the model minimizes the peak passenger queue to $z^* = 6,490$ passengers, occurring at the end of interval 5 (60–75 minutes after the match). The queue is fully cleared by interval 11, well within the 3-hour planning horizon.

Peak Queue (z^*) 6,490 <i>passengers — Interval 5</i>	Queue Cleared Interval 11 <i>165 min post-match</i>	Total Queue Burden 45,290 <i>passenger-intervals</i>	Avg. Wait Time ~28.3 min <i>per passenger</i>
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5.1 Interval-by-Interval Dispatch Plan

Table 1 presents the complete base case solution. The optimizer dispatches the maximum 18 buses in most intervals, with reductions in intervals 4 and 8 driven entirely by the fleet turnaround constraint, not by low demand.

Interval	Demand	Rail Cap.	Buses Disp.	Bus Cap. Added	Total Cap.	Pass. Served	Queue End
1	4,320	1,300	18	990	2,290	2,290	2,030
2	4,080	1,300	18	990	2,290	2,290	3,820
3	3,600	1,300	18	990	2,290	2,290	5,130
4	2,880	1,300	6	330	1,630	1,630	6,380
5	2,400	1,300	18	990	2,290	2,290	6,490
6	1,920	1,300	18	990	2,290	2,290	6,120
7	1,440	1,300	18	990	2,290	2,290	5,270
8	1,200	1,300	6	330	1,630	1,630	4,840
9	960	1,300	18	990	2,290	2,290	3,510
10	480	1,300	18	990	2,290	2,290	1,700
11	480	1,300	16	880	2,180	2,180	0
12	240	1,300	0	0	1,300	240	0

Gold row = peak queue (interval 5, $z^* = 6,490$). **Red rows** = turnaround constraint binding (intervals 4 and 8).

5.2 Queue Dynamics — Key Findings

- **Queue Build Phase (Intervals 1–5):** The queue builds for the first five intervals because outbound demand exceeds combined service capacity. This gap is structural and unavoidable under base case parameters.
- **Turnaround Constraint (Intervals 4 & 8):** Intervals 4 and 8 see dispatch fall from 18 to 6 buses because buses dispatched in the three preceding intervals are still completing their turnaround cycle. This is the most important operational bottleneck in the model.
- **Queue Dissipation Phase (Intervals 6–11):** After interval 5, demand declines and service capacity begins to outpace arrivals, causing the queue to fall steadily until full clearance at interval 11.
- **Rail-Only Clearance (Interval 12):** Interval 12 sees only 240 passengers remaining, entirely within fixed rail capacity. No bus dispatch is required.

The total queue of 45,290 passenger-intervals implies an estimated average waiting burden of approximately 28.3 minutes per passenger ($45,290 \times 15 \text{ min} / 24,000 \text{ passengers}$). This is a rough aggregate; individual waiting times vary substantially.

6. Scenario Analysis — Risk Spectrum

To test model behaviour across a range of plausible operating conditions, three scenarios are defined. Each scenario changes multiple parameters simultaneously, representing coherent planning assumptions rather than isolated parameter changes.

Parameter	Low (Conservative)	Base (Planned)	High (Enhanced)
Total Demand	20,000 pass.	24,000 pass.	28,000 pass.
Rail Capacity (R_t)	1,050 pass./int.	1,300 pass./int.	1,550 pass./int.
Supporting Buses (B)	45 buses	60 buses	70 buses
Bus Capacity (C^b)	45 pass./bus	55 pass./bus	65 pass./bus
Turnaround Time (τ)	75 min ($h=5$)	60 min ($h=4$)	45 min ($h=3$)
Loading Limit (L)	12 buses/int.	18 buses/int.	24 buses/int.
Solver Status	Optimal ✓	Optimal ✓	Optimal ✓
Peak Queue (z^*)	4,945 pass.	6,490 pass.	7,570 pass.
Total Queue (Σq_t)	43,875 pass-int.	45,290 pass-int.	41,810 pass-int.
Total Buses Dispatched	114	172	210

6.1 Conservative Scenario

The low scenario reduces total demand to 20,000 passengers but simultaneously constrains bus operations: 45 buses, 45 passengers per bus, 75-minute turnaround, and a 12-bus loading limit. Despite lower demand, more constrained operations result in a peak queue of 4,945 passengers. The tighter fleet and slower turnaround partially offset the benefit of lower demand, underscoring that operational capability matters as much as passenger volume.

6.2 High Demand — Enhanced Operations Scenario

The high scenario increases demand to 28,000 passengers but simultaneously strengthens every operational dimension: 70 buses, 65-passenger effective capacity, 45-minute turnaround, and a 24-bus loading limit. The peak queue rises to 7,570 passengers — higher than the base case. However, the total queue (41,810 passenger-intervals) is lower than the base case (45,290), reflecting that enhanced operations reduce the aggregate waiting burden even though the instantaneous worst case is larger.

Cross-Scenario Insight

The high scenario is a high-demand, high-capability case — not simply a high-demand case. The improved operational parameters are the primary reason total queue is lower than the base case despite greater demand. This underscores a key managerial principle: operational improvements (especially faster turnaround) can have a disproportionate effect on total passenger waiting time, even when peak queue levels remain elevated.

7. Demand Sensitivity Analysis — Capacity Risk Register

This analysis quantifies how the model responds to varying levels of total outbound passenger demand while holding all base-case operating parameters fixed. This isolates demand as a standalone risk driver, providing planners with a full risk spectrum beyond the three-scenario framework.

Total Demand	Solver Status	Peak Queue (z^*)	Total Queue (Σq_t)	Buses Dispatched	Turnaround (h)
18,000	Optimal	2,660	11,060	114	4
20,000	Optimal	3,900	20,320	138	4
22,000	Optimal	5,140	31,810	156	4
24,000	Optimal ★	6,490	45,290	172	4
26,000	Optimal ⚠	7,930	61,380	180	4
28,000	Optimal ⚠	9,370	79,080	180	4
30,000	Optimal ⚠	10,920	96,780	180	4
32,000	Optimal ⚠	12,520	114,480	180	4

★ = Base case. ⚠ = Operational ceiling reached (180 total dispatches). Additional demand is entirely unabsorbed by the bus system.

Capacity Ceiling — Critical Risk Threshold

At 26,000 passengers and above, the model exhausts its 180 total bus dispatch capacity (the maximum achievable under base fleet and turnaround constraints). Beyond this ceiling, additional demand is entirely unabsorbed during the peak period. For a World Cup Final that could attract 28,000–32,000 transit passengers, base-case operational parameters are

insufficient. Planners must expand fleet, reduce turnaround time, increase rail capacity, or implement demand management strategies before reaching this threshold.

8. One-at-a-Time Sensitivity Analysis

Because scenario analysis changes multiple parameters simultaneously, this section conducts a one-at-a-time (OAT) analysis. Each parameter is varied while all others are held at base-case values, isolating the independent effect on the optimized peak queue z^* .

Parameter	Test Range	Best z^*	Worst z^*	Best Δ	Worst Δ	Rank
Total Demand (D)	18,000–30,000	2,660	10,920	–3,830	+4,430	#1
Rail Capacity (R_t)	900–1,700 pass.	4,780	8,520	–1,710	+2,030	#2
Bus Capacity (C^b)	35–75 pass./bus	5,180	8,050	–1,310	+1,560	#3
Fleet Size (B)	30–90 buses	5,830	8,240	–660	+1,750	#4
Turnaround Time (τ)	30–90 min	5,830	8,100	–660	+1,610	#5
Loading Limit (L)	10–24 buses/int	6,380	8,100	–110	+1,610	#6

Base-case $z^* = 6,490$ passengers. Negative Δ = improvement (queue reduction). Positive Δ = deterioration (queue increase).

8.1 Key Parameter Insights

- **Rank 1 — Total Demand:** Total Demand is the strongest driver of peak queue by a wide margin. The sensitivity range spans 8,260 passengers (from –3,830 best-case to +4,430 worst-case). Even with identical operational parameters, higher passenger volume quickly pushes the system toward extreme congestion.
- **Rank 2 — Rail Capacity:** Rail capacity is the second most important factor. Improving rail capacity from 1,300 to 1,700 passengers per interval reduces peak queue by 1,710 passengers. Reducing it to 900 increases the peak by 2,030 passengers. This asymmetry indicates the system is more vulnerable to rail degradation than it benefits from improvement.
- **Rank 3 — Bus Capacity:** Each extra passenger per bus scales with total buses dispatched, yielding a strong multiplier effect (–1,310 / +1,560 range).
- **Ranks 4 & 5 — Fleet & Turnaround:** Fleet size and turnaround time have symmetric best-case improvements (–660 each) but asymmetric worst-case effects (+1,750 and +1,610). Deterioration hurts more than improvement helps, because other constraints become binding when bus-side conditions improve.
- **Rank 6 — Loading Limit:** Loading limit has the smallest improvement effect (–110 passengers), confirming that once L exceeds the fleet's effective dispatch rate, it is no longer the binding constraint.

8.2 Two-Way Sensitivity: Fleet Size × Turnaround Time

The two-way heatmap (Figure 8 in the original analysis) reveals that faster turnaround can partially compensate for a smaller fleet. For example, 30 buses with a 30-minute turnaround achieves the same $z^* = 6,490$ as the base case (60 buses, 60-minute turnaround). Diminishing returns are also evident: multiple cells with larger fleets all produce $z^* = 5,830$, indicating that rail capacity, loading limit, or demand timing has become the binding constraint. Planners must identify which constraint is actually binding before investing in fleet expansion.

Project Management Implication

The OAT and two-way sensitivity results provide a constraint hierarchy for resource prioritisation. Before committing capital to fleet expansion, planners should verify whether turnaround reduction, rail capacity improvement, or demand management would deliver greater queue reduction per dollar invested. The sensitivity analysis provides the quantitative basis for that prioritisation decision.

9. Managerial Recommendations

The following recommendations are grounded in the model's results and the structural properties of the MILP formulation. They are presented as actionable directives for NJ TRANSIT planners, stadium operations managers, and transportation decision-makers responsible for the 2026 World Cup egress plan.

#	Recommendation	Model Evidence	Operational Action
	Front-Load Bus Dispatch	Optimizer assigns maximum 18 buses in intervals 1–3. Any delay causes queue accumulation that must be worked off over subsequent intervals, raising the peak.	Pre-position buses in the staging area before the match concludes. Treat the first 30–45 minutes after the final whistle as the critical deployment window.
	Prioritise Turnaround Time Reduction	Fleet turnaround is a binding constraint in intervals 4 and 8, reducing dispatch from 18 to 6 buses. Reducing τ from 60 to 45 min increases effective reuse rate without adding buses.	Implement dedicated return lanes, multiple-door loading, coordinated bus movements at Secaucus, and pre-staged return buses to minimise dwell time between trips.
	Implement Demand Management Strategies	The model treats demand as fixed, but a 10% shift of first-interval demand (≈ 430 passengers) to later intervals directly reduces the early-interval queue that drives the peak.	Deploy staggered egress by stadium section, on-site post-match programming, and real-time crowd-flow guidance via the stadium app to spread departure timing.
	Plan Contingencies Above the Capacity Ceiling	At 26,000+ passengers, the model exhausts 180 total dispatches. Additional demand is entirely	Define a maximum acceptable queue threshold. If predicted peak exceeds it, activate pre-approved contingency procedures: additional buses,

	unabsorbed under base operating parameters.	adjusted dispatch rates, or coordinated entry slowdown at the egress zone.
Address the Most Binding Constraint First	OAT analysis produces a clear sensitivity ranking: Demand > Rail > Bus Capacity > Fleet/Turnaround > Loading Limit.	Use the sensitivity ranking as a capital allocation guide. Verify which constraint is binding before investing in fleet expansion — rail capacity or turnaround reduction may deliver greater ROI.
Establish Acceptable Queue Thresholds	The model identifies $z^* = 6,490$ under base case. Whether this is operationally acceptable depends on platform capacity and crowd-management standards outside model scope.	Define the maximum queue level that can be safely and comfortably managed at the egress area. Use this threshold to trigger contingency protocols during real-time operations.

10. Limitations & Future Scope

10.1 Current Model Limitations

- **Secaucus Downstream Capacity:** The receiving end at Secaucus Junction is treated as unconstrained. In practice, platform congestion at Secaucus could increase effective turnaround time and reduce fleet utilization.
- **Fixed Turnaround Assumption:** Turnaround time is treated as a fixed parameter rather than a traffic-dependent variable. Real turnaround time will vary based on road congestion, boarding speed, and weather conditions.
- **Demand Profile Uncertainty:** Outbound demand profiles are scenario-based planning assumptions, not empirically derived from interval-level egress data. The actual departure curve will vary by match result, weather, kickoff time, and individual post-match activity choices.
- **Single-Mode Focus:** The model covers only NJ TRANSIT-related service. Private car, rideshare, and charter bus modes are excluded. If higher-than-expected share of spectators chooses NJ TRANSIT, actual demand could exceed even the high-demand scenario.

10.2 Recommended Future Extensions

- **Data Calibration:** Integrate interval-level boarding and passenger-count data from comparable large events at MetLife Stadium to calibrate demand profiles empirically.
- **Downstream Modelling:** Extend the model to incorporate Secaucus Junction downstream capacity as an explicit constraint.
- **Stochastic Demand:** Develop stochastic demand profiles to generate risk-adjusted solutions with probability-weighted queue distributions.
- **Multi-Mode Extension:** Build joint optimization across multiple egress modes (rail, bus, rideshare, charter) for a system-wide perspective.
- **Dynamic Re-Optimisation:** Implement real-time dynamic re-optimization based on observed passenger counts and live bus GPS locations.

11. Conclusion

This project develops and applies a time-indexed Mixed-Integer Linear Programming model to the post-match bus dispatch scheduling problem at MetLife Stadium during the FIFA World Cup 2026. The model optimizes bus dispatch across 12 consecutive 15-minute intervals, minimizing the worst-case passenger queue during the 3-hour post-match period.

The base case achieves an optimal peak queue of 6,490 passengers, with the queue fully cleared by interval 11 and an estimated average waiting burden of 28.3 minutes per passenger. Fleet turnaround time is identified as the critical operational bottleneck — the constrained intervals (4 and 8) are driven by the rolling-window fleet availability constraint, not by low demand.

Demand sensitivity analysis reveals a capacity ceiling at approximately 26,000 passengers, above which the bus system exhausts its dispatch capacity and additional demand goes unabsorbed. One-at-a-time sensitivity analysis confirms that total demand is the strongest single driver of peak queue, followed by rail capacity, bus capacity, fleet size, and turnaround time. The two-way fleet–turnaround heatmap shows that a smaller fleet with a faster turnaround can match the performance of a larger fleet — an important insight for operational planning under resource constraints.

Project Conclusion

This model is best interpreted as a practical decision-support tool for pre-event planning. The most valuable next step is to collect interval-level boarding and bus-movement data from comparable large events at MetLife Stadium to calibrate demand profiles more precisely and validate model predictions against observed post-match operations. All six managerial recommendations should be reviewed by NJ TRANSIT planners in advance of the July 19, 2026 Final.

References

1. FIFA. (n.d.). FIFA World Cup 2026™ venues.
<https://www.fifa.com/en/tournaments/mens/worldcup/canadamexicousa2026/articles/venues>
2. MetLife Stadium. (2024, February 9). MetLife Stadium selected as host venue for FIFA World Cup 26 final. <https://www.metlifestadium.com/news/detail/metlife-stadium-selected-as-host-venue-for-fifa-world-cup-26-final>
3. NJ TRANSIT. (n.d.). New multilevel rail cars dedicated to 9 New Jersey communities.
<https://www.njtransit.com/press-releases/new-multilevel-rail-cars-dedicated-9-new-jersey-communities>
4. NJ TRANSIT. (n.d.). NJ TRANSIT board approves purchase of new rail cars.
<https://www.njtransit.com/press-releases/nj-transit-board-approves-purchase-new-rail-cars>
5. NJ TRANSIT. (n.d.). Ride NJ TRANSIT to the World's Game! <https://www.njtransit.com/fifa>
6. NJ TRANSIT. (2025, May 29). NJ TRANSIT offers rail and bus service for FIFA Club World Cup games at MetLife Stadium. <https://www.njtransit.com/press-releases/nj-transit-offers-rail-and-bus-service-fifa-club-world-cup-games-metlife-stadium>
7. NJ TRANSIT. (2025, July 17). Board meeting minutes (Open Session).
https://content.njtransit.com/sites/default/files/board/meeting_minutes/2025_07_17_OpenSess.pdf
8. NJ TRANSIT. (2025, September 19). Board meeting minutes (Open Session).
https://content.njtransit.com/sites/default/files/board/meeting_minutes/2025_09_19_OpenSess.pdf